

1 Exam Practice

Question 1.1 (Practice 1). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with compact support. Let $E \in \mathbb{R}$ and $\eta > 0$. Compute,

$$\lim_{\eta \rightarrow 0^+} \operatorname{Im} \left[\int_{\mathbb{R}} \frac{f(x)}{x - (E + i\eta)} dx \right]$$

If $\eta > 0$ show that the function

$$E \rightarrow \operatorname{Im} \left[\int_{\mathbb{R}} \frac{f(x)}{x - (E + i\eta)} dx \right]$$

is smooth.

Proof. Note that the function in the integral is

$$\operatorname{Im} \left(\frac{f(x)}{x - (E + i\eta)} \right) = \operatorname{Im} \left(\frac{(x - E + i\eta)f(x)}{(x - E)^2 + \eta^2} \right) = \frac{\eta f(x)}{(x - E)^2 + \eta^2}$$

Since f is continuous with compact support, we have f is only nonzero on $[-N, N]$ and $|f(x)| \leq M$ on this interval. We then get

$$\left| \operatorname{Im} \left[\int_{\mathbb{R}} \frac{f(x)}{x - (E + i\eta)} dx \right] \right| \leq \int_{-N}^N \frac{\eta M dx}{(x - E)^2 + \eta^2} \leq \int_{-N}^N \frac{M dx}{\eta} = \frac{2NM}{\eta} \rightarrow 0 \text{ as } \eta \rightarrow 0^+$$

This proves the first part.

Let F be that second function. Then we have

$$F(E) = \int_{\mathbb{R}} \frac{\eta f(x)}{(x - E)^2 + \eta^2} dx = f * g(E)$$

where $g(t) = \frac{\eta}{t^2 + \eta^2}$. Note that g is smooth. It thus suffices to show that $F' = f * g'$ by repeated application of this. For any E, h , we have

$$\begin{aligned} \left| \frac{F(E + h) - F(E)}{h} - f * g'(E) \right| &= \left| \frac{1}{h} \int_{\mathbb{R}} \frac{\eta f(x)}{(x - E - h)^2 + \eta^2} dx - \frac{1}{h} \int_{\mathbb{R}} \frac{\eta f(x)}{(x - E)^2 + \eta^2} dx - \int_{\mathbb{R}} g'(E - x) f(x) dx \right| \\ &\leq 2MN \sup_{x \in [-N, N]} \left| \frac{g(E - x + h) - g(E - x)}{h} - g'(E - x) \right| \rightarrow 0 \text{ as } h \rightarrow 0 \end{aligned}$$

The latter limit is uniform since g is C^1 . Hence F is smooth. □

Question 1.2 (Practice 2). Show that a monotonic function $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable.

Proof. Without loss of generality, suppose f is increasing. It suffices to show that for any $(a, b) \subseteq \mathbb{R}$, $f^{-1}(a, b)$ is Borel measurable. If it is empty or a singleton then we are done. Otherwise, we show that $f^{-1}(a, b)$ is an interval. For that, we let $x < y \in f^{-1}(a, b)$ and show that any $t \in (x, y)$ is in $f^{-1}(a, b)$. Since f is increasing, we have $f(x) \leq f(t) \leq f(y)$, so $f(t) \in [f(x), f(y)] \subseteq (a, b)$, and hence $t \in f^{-1}(a, b)$. Therefore, f is Borel measurable. □

Question 1.3 (Practice 3). If $m(E) = 0$ what can you say about $L^p(E)$?

Proof. Note that any set of outer measure zero is measurable, so any function $E \rightarrow \overline{\mathbb{R}}$ is measurable. Furthermore, for any such f , we have

$$\int_E |f|^p = 0 \Rightarrow f = 0$$

This shows that $L^p(E)$ contains only the zero function (under the equivalence relation). □

Question 1.4 (Practice 4). For which p are trigonometric polynomials dense in $L^p(-\pi, \pi)$?

Proof. Recall that functions in $L^p(-\pi, \pi)$ can be approximated by continuous functions on $(-\pi, \pi)$ of bounded support. If we can show that each continuous function on $[a, b] \subseteq (-\pi, \pi)$ can be approximated by trigonometric polynomials, then we will have our result. One can use a Stone-Weierstrass Theorem argument to show that this can be done with the uniform convergence topology, so we just need to show that uniform convergence on $[a, b]$ implies L^p convergence on $(-\pi, \pi)$, given that the functions are zero a.e. outside $[a, b]$. Suppose continuous $f_n : (-\pi, \pi) \rightarrow \mathbb{R}$ converge to a continuous f uniformly on $[a, b]$ and f, f_n are supported in $[a, b]$. Then, we have

$$\int_{(-\pi, \pi)} |f_n - f|^p \leq (b - a) \sup_{x \in [a, b]} |f_n(x) - f(x)|^p \rightarrow 0$$

as $n \rightarrow \infty$, which completes the proof. $\square$

Question 1.5 (Practice 5). Let (M, d) be a metric space and for $x \in M$ and a nonempty set A define

$$d(x, A) := \inf_{y \in A} d(x, y)$$

Prove that $d(x, A)$ is a continuous function of x . Prove that A is closed iff the following statement holds: for all $x \in M$ we have that $d(x, A) = 0$ iff $x \in A$.

Proof. Suppose A is closed. Clearly if $x \in A$, we have $d(x, A) = 0$. If $x \notin A$, then since A is closed, x is an interior point of the open set $M \setminus A$, so there is some $r > 0$ such that $B_x(r) \subseteq M \setminus A$. This means that every point $y \in A$ satisfies $d(x, y) \geq r$, hence $d(x, A) \geq r > 0$.

Conversely, suppose that $d(x, A) = 0$ iff $x \in A$. We show that A is closed by showing that its complement is open. For any $x \in M \setminus A$, we let $r = d(x, A) > 0$. For any $y \in B_x(r)$, we have $d(x, y) < r = d(x, A) = \inf_{z \in A} d(x, z)$, so y cannot be in A . Hence, $y \in M \setminus A \Rightarrow B_x(r) \subseteq M \setminus A$, and therefore, A is closed. $\square$

Question 1.6 (Practice 6). Let $f_n(x)$ be a sequence of polynomials and let $f(x) = \lim_{n \rightarrow \infty} f_n(x)$. Is f necessarily continuous?

Proof. Not necessarily. Consider the space $C[0, 1]$ and the polynomials $f_n(x) = x^n$. For $x \in [0, 1)$, $f(x) = \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} x^n = 0$, and $f(1) = \lim_{n \rightarrow \infty} 1 = 1$. Clearly, f is not continuous. $\square$

Question 1.7 (Practice 7). Let (M, d) be metric space. We say a sequence $x_n \in M$ is fast-Cauchy if there is a summable sequence $a_n > 0$ (i.e., $\sum a_n < \infty$) such that $d(x_n, x_{n+1}) < a_n$. Disprove or prove the following:

- (a) A fast-Cauchy sequence is Cauchy
- (b) Every Cauchy sequence has a fast-Cauchy subsequence
- (c) A sequence is Cauchy iff every subsequence has a subsequence that is fast-Cauchy

Proof. (a). This one is true. Note that $\sum_{k=n}^{\infty} a_k = \sum_{k=0}^{\infty} a_k - \sum_{k=0}^{n-1} a_k \rightarrow 0$ as $n \rightarrow \infty$, so given $\varepsilon > 0$, we can find $N > 0$ such that $\sum_{k=n}^{\infty} a_k < \varepsilon$. Then, for any $m > n \geq N$, we have

$$d(x_n, x_m) \leq \sum_{k=n}^{m-1} d(x_k, x_{k+1}) < \sum_{k=n}^{m-1} a_k \leq \sum_{k=n}^{\infty} a_k < \varepsilon$$

Hence x_n is a Cauchy sequence.

(b). This one is also true. For every $k > 0$, since x_n is Cauchy, we can choose an index $n_k > n_{k-1}$ such that for all $j \geq n_k$, we have $d(x_j, x_{j+1}) < 2^{-k}$. By construction, $d(x_{n_k}, x_{n_{k+1}}) < 2^{-k}$, and $a_k = 2^{-k}$ is a geometric series, hence is summable. Thus x_n has a fast Cauchy subsequence.

(c). This one is false if M has at least two points $p \neq q$. Let x_n be the sequence $x_{2k} = p, x_{2k+1} = q$. Clearly, this is not Cauchy. However, any subsequence of it has a further subsequence that is either constant p or constant q , which is clearly fast Cauchy. $\square$